Lab Sheet 01 – Java Basics

Objective

Understand basic Java programming concepts including printing output, variables, data types, loops, array, conditional statements and user input.

Question 1: School Performance Report System

Scenario

A school wants a simple system to analyze student performance for a class test.You are asked to develop a Java program to help the teacher generate a report.

Task

Write a program that:

1. Inputs the number of students **n**
2. For each student:
    - Enter **student name**
    - Enter **marks (0–100)**
3. Store the data using arrays

The program should then:

- Calculate the **average marks of the class**
- Find the **highest mark scored**
- Categorize students into:
    - **Distinction** (marks ≥ 75)
    - **Pass** (marks 50–74)
    - **Fail** (marks < 50)

Output Format

Display the report as:

--- Class Report ---
Average Marks: X
Highest Marks: Y
Distinction Students: A
Pass Students: B
Fail Students: C

Requirements

- Use **arrays**
- Use **loops**
- Use **if-else conditions**
- Use at least **one method** (for average or max calculation)

Question 2: ATM Machine Simulation

Scenario

You are developing a simple software simulation for an ATM machine used in a bank.

The system should allow a user to perform basic banking operations.

Task

Write a Java program that:

1. Accepts an **initial account balance**
2. Displays the following menu repeatedly:

    1 → Deposit Money
    2 → Withdraw Money
    3 → Check Balance
    4 → Exit

3. Performs operations based on user choice

Rules

- Deposit → add money to balance
- Withdraw → subtract money only if sufficient balance
- If insufficient balance → display message
- Continue until user selects **Exit**

Requirements

- Use **do-while loop**
- Use **switch-case**
- Use **methods** (deposit, withdraw)
- Use user input

Sample Interaction

Enter balance: 1000
1.Deposit 2.Withdraw 3.Check Balance 4.Exit
Choice: 1
Enter amount: 500
Balance updated

Choice: 3
Balance: 1500